

D881

Advanced Zoology and Botany with Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher is preparing a virtual plant lab for her third-period class, which meets right after lunch. The simulation lets students change conditions around a leafy bean plant that sits in a blue clay pot. Students coat the undersides of all leaves with petroleum jelly, which blocks the stomata. What should they predict about water movement up the stem?

- A. Water movement stays the same because roots pump water upward regardless of the leaves.
- B. Water movement up the xylem slows sharply because transpiration from the leaves is blocked.
- C. Water movement speeds up because water has nowhere to escape and builds pressure.
- D. Sugar movement in the xylem stops, but water is unaffected.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A biology club at an arts-focused high school measures CO₂ uptake by a jade plant, a desert succulent, in a sealed chamber. The plant was donated by a local garden center and is about four years old. Readings are taken every hour for 24 hours. When would students expect the highest CO₂ uptake?

- A. Evenly across all 24 hours.
- B. At midday, when light is strongest.
- C. At midday, because CO₂ is fixed in separate bundle-sheath cells.
- D. At night, when the stomata are open and CO₂ is stored as organic acids.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

In a zoology unit, students compare a trout from a cold mountain stream with a model of its body fluids. The trout is about 30 cm long and was raised in a state hatchery. The stream water is far less salty than the trout's blood. Which pair of adaptations fits this trout?

- A. It needs no special adaptations because cold water slows osmosis.
- B. It drinks very little and excretes large amounts of dilute urine.
- C. It drinks large amounts of water and excretes small amounts of concentrated urine.
- D. It stores extra urea in its blood to match the stream water.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A marine bony fish lives in seawater that is saltier than its body fluids. Which statement describes how it keeps water balance?

- A.** It drinks almost nothing and makes large amounts of dilute urine.
- B.** It never drinks, because water enters its body through the skin.
- C.** It drinks seawater and actively excretes salt through its gills.
- D.** It absorbs salt through its gills to match the seawater.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

During a field study at a wildlife refuge, students compare a shrew (about 5 g) with published data for a caribou (about 100 kg). Both live in the same cold tundra region and have thick fur. Which animal has a higher metabolic rate per gram of body mass, and why?

- A.** The shrew, because its high surface-area-to-volume ratio makes it lose heat faster per gram.
- B.** Both are equal, because they live in the same climate.
- C.** The caribou, because it has more total skin to lose heat from.
- D.** The caribou, because larger animals always need more total food.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A student in the school's greenhouse club grows oat seedlings in a box with one small window on the left side. The seeds are an heirloom variety and are watered every morning with rainwater. After two days the shoots bend to the left. Which explanation should the teacher accept?

- A.** Rainwater lowered the pH on the left side of the stems.
- B.** Auxin moved to the shaded right side, so cells on that side grew longer and the shoot bent left.
- C.** Auxin built up on the lit left side, so those cells grew faster.
- D.** Cells on the left died from too much light.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A class builds a cladogram of five vertebrates using a lamprey as the outgroup. One student spent the weekend at an aquarium and brings photos. The trait list includes a backbone, jaws, and four limbs. Which trait gives evidence that a frog and a lizard are in a clade that excludes a shark?

- A. Similar body size
- B. A backbone
- C. Four limbs
- D. Jaws

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Two insect species both have wings, and so do all their close relatives and the outgroup used in the study. What does sharing wings tell you about whether these two species form a clade?

- A. Nothing; a trait also found in the outgroup is ancestral and does not define a smaller clade.
- B. They must be in different clades.
- C. They must be each other's closest relatives.
- D. Wings are a synapomorphy for the pair.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A food web poster made for back-to-school night shows grasses capturing 20,000 kcal per square meter per year. The poster was printed in full color and hangs near the science wing entrance. Using a 10% transfer rule, about how much energy reaches the hawks that eat snakes that eat mice that eat the grass seeds?

- A. 200 kcal
- B. 18,000 kcal
- C. 2,000 kcal
- D. 20 kcal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Hawks at the top of a four-level food chain (plants, mice, snakes, hawks) receive 15 kcal per square meter per year. Using the 10% rule, how much energy did the plants capture?

- A. 0.015 kcal
- B. 1,500 kcal
- C. 150 kcal
- D. 15,000 kcal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Students in a zoology lab test whether temperature affects the heart rate of Daphnia. They use a microscope that the school bought last year and a timer app on a tablet. They place Daphnia in water at 10, 15, 20, and 25 degrees C and count heartbeats per minute. What is the dependent variable?

- A. Heart rate in beats per minute
- B. The species of Daphnia
- C. Water temperature
- D. The type of microscope

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A student wants to test whether light color affects plant growth. She puts one plant under red light with fertilizer and another under blue light without fertilizer. What is the main flaw in this design?

- A. She should measure light color instead of growth.
- B. A test with only two plants can never give any useful data.
- C. Two independent variables change, so any difference cannot be linked to light color.
- D. There is no dependent variable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A teacher is preparing a virtual plant lab ~~for her third period class, which meets right after lunch.~~ The simulation lets students change conditions around a leafy bean plant ~~that sits in a blue clay pot.~~ Students coat the undersides of all leaves with petroleum jelly, which blocks the stomata. What should they predict about water movement up the stem?

Target word: effect on water transport **Decides:** Blocked stomata stop transpiration, which is the pull that lifts water.

A	Adjacent category	Root pressure is minor; the main driver is transpiration pull.
B	Correct	Evaporation at the leaves creates the tension that pulls water up; blocking stomata removes most of that pull.
C	Inverted operation	Reverses the mechanism; xylem flow is pulled from above, not pushed by trapped water.
D	Confusion set	Sugar moves in phloem, not xylem; mixes up the two vascular tissues.

2. Answer D Concept 2

A biology club ~~at an arts-focused high school~~ measures CO₂ uptake by a jade plant, a desert succulent, in a sealed chamber. ~~The plant was donated by a local garden center and is about four years old.~~ Readings are taken every hour for 24 hours. When would students expect the highest CO₂ uptake?

Target word: time of CO₂ uptake **Decides:** Jade is a CAM plant, which opens stomata at night.

A	Absolute language	CAM uptake is strongly concentrated at night, not constant.
B	Inverted operation	That is the C ₃ pattern; a CAM plant keeps stomata closed in the hot day.
C	Confusion set	That describes C ₄ (separation in space), not CAM (separation in time).
D	Correct	CAM plants take in CO ₂ at night to reduce water loss, then use it in the Calvin cycle by day.

3. Answer B Concept 3

In a zoology unit, students compare a trout from a ~~cold mountain~~ stream with a model of its body fluids. ~~The trout is about 30 cm long and was raised in a state hatchery.~~ The stream water is far less salty than the trout's blood. Which pair of adaptations fits this trout?

Target word: freshwater fish osmoregulation **Decides:** Surrounding water is less salty than body fluids, so water moves in.

A	Camouflage	Temperature detail does not decide the direction of water movement.
B	Correct	Water enters by osmosis, so the fish must get rid of water and keep salts.
C	Inverted operation	That is the marine bony fish strategy, which loses water to salty surroundings.
D	Confusion set	Urea retention is the shark (cartilaginous fish) strategy in seawater.

4. Answer C Concept 3 • transfer

A marine bony fish lives in seawater that is saltier than its body fluids. Which statement describes how it keeps water balance?

Target word: marine osmoregulation **Decides:** Water leaves the body by osmosis into saltier seawater.

A	Inverted operation	That is the freshwater strategy, where water floods in.
B	Absolute language	Water leaves, not enters, a marine bony fish.
C	Correct	It must replace lost water, so it drinks and then pumps out the extra salt.
D	Inverted operation	Absorbing salt would worsen water loss; it excretes salt.

5. Answer A Concept 4

~~During a field study at a wildlife refuge,~~ students compare a shrew (about 5 g) with published data for a caribou (about 100 kg). Both live in the same cold tundra region ~~and have thick fur.~~ Which animal has a higher metabolic rate per gram of body mass, and why?

Target word: per-gram metabolic rate **Decides:** Smaller body means larger surface-area-to-volume ratio and faster heat loss.

A	Correct	Small animals lose relatively more heat, so they burn more energy per gram to stay warm.
B	Camouflage	Shared climate does not make per-gram rates equal.
C	Inverted operation	Total surface is larger, but surface per unit volume is much smaller.
D	Adjacent category	True for total energy, but the question asks per gram.

6. Answer B Concept 5

~~A student in the school's greenhouse club grows oat seedlings in a box with one small window on the left side. The seeds are an heirloom variety and are watered every morning with rainwater.~~ After two days the shoots bend to the left. Which explanation should the teacher accept?

Target word: mechanism of bending **Decides:** Auxin moves to the shaded side and cells there elongate.

A	Camouflage	Watering detail does not decide the direction of bending.
B	Correct	Uneven elongation, driven by auxin on the shaded side, bends the shoot toward light.
C	Inverted operation	Auxin collects on the shaded side, not the lit side.
D	Adjacent category	Phototropism is growth, not cell death.

7. Answer C Concept 6

A class builds a cladogram of five vertebrates using a lamprey as the outgroup. ~~One student spent the weekend at an aquarium and brings photos.~~ The trait list includes a backbone, jaws, and four limbs. Which trait gives evidence that a frog and a lizard are in a clade that excludes a shark?

Target word: which trait defines the frog-lizard clade **Decides:** Four limbs are derived and shared by frog and lizard, not shark.

A	Confusion set	Overall appearance is not a derived character used in cladistics.
B	Adjacent category	All vertebrates, including the shark and lamprey, share it; it is ancestral here.
C	Correct	Four limbs is a shared derived trait of tetrapods that sharks lack.
D	Dropped element	Sharks also have jaws, so jaws cannot separate them from the frog and lizard.

8. Answer A Concept 6 • transfer

Two insect species both have wings, and so do all their close relatives and the outgroup used in the study. What does sharing wings tell you about whether these two species form a clade?

Target word: value of an ancestral trait **Decides:** The trait is shared with the outgroup, so it is ancestral here.

A	Correct	Only shared derived traits support grouping.
B	Inverted operation	The trait gives no evidence either way, not evidence against.
C	Absolute language	Shared ancestral traits cannot establish close relationship.
D	Confusion set	A synapomorphy is derived relative to the outgroup; this trait is not.

9. Answer D Concept 7

A food web poster ~~made for back-to-school night~~ shows grasses capturing 20,000 kcal per square meter per year. ~~The poster was printed in full color and hangs near the science wing entrance.~~ Using a 10% transfer rule, about how much energy reaches the hawks that eat snakes that eat mice that eat the grass seeds?

Target word: energy at the fourth trophic level **Decides:** Three transfers: grass to mice to snakes to hawks.

A	Dropped element	Stops at the snakes, one transfer short.
B	Inverted operation	Subtracts 10% instead of keeping 10%.
C	Wrong stem number	Only one transfer applied.
D	Correct	$20,000 \times 0.1 \times 0.1 \times 0.1 = 20$.

10. Answer D Concept 7 • transfer

Hawks at the top of a four-level food chain (plants, mice, snakes, hawks) receive 15 kcal per square meter per year. Using the 10% rule, how much energy did the plants capture?

Target word: solve for producer energy **Decides:** Reverse three 10% transfers by multiplying by 10 three times.

A	Inverted operation	Multiplies by 0.1 going down instead of by 10.
B	Dropped element	Only two transfers reversed.
C	Dropped element	Only one transfer reversed.
D	Correct	$15 \times 10 \times 10 \times 10 = 15,000$.

11. Answer A Concept 8

Students in a zoology lab test whether temperature affects the heart rate of Daphnia. ~~They use a microscope that the school bought last year and a timer app on a tablet.~~ They place Daphnia in water at 10, 15, 20, and 25 degrees C and count heartbeats per minute. What is the dependent variable?

Target word: which variable is measured **Decides:** Heartbeats are what is counted in response.

A	Correct	It is measured and depends on the temperature students set.
B	Adjacent category	That is a controlled variable held constant.
C	Inverted operation	That is the independent variable the students change.
D	Camouflage	Equipment detail is not a variable in the design.

12. Answer C Concept 8 · transfer

A student wants to test whether light color affects plant growth. She puts one plant under red light with fertilizer and another under blue light without fertilizer. What is the main flaw in this design?

Target word: identify the design flaw **Decides:** Two variables change at once.

A	Inverted operation	Swaps the independent and dependent variables.
B	Absolute language	Small sample size is a limit, but the key flaw is the confound.
C	Correct	Fertilizer is a confounding variable; it must be held constant.
D	Dropped element	Plant growth is the dependent variable.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	12	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Transpiration pulls water up the xylem	Water rises in xylem because evaporation from leaves creates tension that pulls a cohesive water column up; phloem moves sugars from source to sink.	Thinking roots pump water up, or that phloem carries water from roots to leaves.	A question about what drives water upward, or wilting when stomata close or humidity changes.
2	CAM and C4 photosynthesis	CAM plants open stomata at night and store CO ₂ as acids to save water; C4 plants separate CO ₂ fixation in space to limit photorespiration.	Assuming all plants take in CO ₂ during daylight, or mixing CAM (time) with C4 (space).	A desert succulent, night-time CO ₂ uptake, or hot dry conditions.
3	Osmoregulation in fish	Freshwater fish gain water by osmosis, so they drink little and excrete large amounts of dilute urine; marine bony fish lose water, so they drink seawater and excrete salt.	Giving the freshwater and marine strategies the reverse of each other.	The salt concentration of the water compared with the fish's body fluids.
4	Surface-area-to-volume and heat loss	Smaller bodies have more surface per unit volume, so they lose heat faster and need higher mass-specific metabolic rates.	Thinking larger animals lose heat faster because they have more total skin.	Two animals of different size in the same cold climate, or per-gram metabolic rate.
5	Auxin and phototropism	Auxin moves to the shaded side of a shoot tip, cells there elongate more, and the shoot bends toward the light.	Thinking auxin builds up on the lit side or that light cells grow faster.	Light from one side; covering or removing the tip.
6	Cladistics uses shared derived traits	Clades are defined by shared derived characters (synapomorphies); shared ancestral traits do not show close relationship.	Grouping organisms by any shared trait, including ancestral ones, or by overall appearance.	Which trait supports grouping two taxa; an outgroup that also has the trait.
7	Trophic efficiency	Only about 10% of energy at one trophic level is passed to the next; the rest is lost as heat and used in life processes.	Assuming most energy passes up the chain, or that energy is recycled like matter.	Energy values at producers and asked for a higher trophic level.
8	Controlled experiment design	Change one independent variable, measure the dependent variable, hold other variables constant, and include a control for comparison.	Confusing the variable that is changed with the one that is measured, or changing two things at once.	A lab plan that lists what is changed, what is measured, and what is kept the same.

Course scope (public description): Advanced Zoology & Botany with Lab is an in-depth exploration of advanced topics in zoology and botany, including the study of animal and plant physiology, ecology, evolution, and taxonomy. This course will engage students in virtual, hands-on laboratory exercises to develop practical skills in virtual or hands-on lab activities, data collection, and analysis. This course builds on concepts from General Biology I and II, and Evolutionary Biology and contains instructional teaching strategies for classroom delivery.